

What are Azure Service Models?

Answer:

- IaaS → Infrastructure (VMs, Network)
- PaaS → Platform (App Service, DB)
- SaaS → Software (Office 365)
- Control decreases from IaaS → SaaS

One-liner:

Azure service models define levels of control: IaaS (full), PaaS (partial), SaaS (minimal).

What is Azure Active Directory?

Answer :

- Identity management service
- Provides authentication & authorization
- Supports SSO
- Integrates with apps

One-liner:

Azure AD manages user identity and access securely.

What is NSG?

Answer :

- Network Security Group
- Controls traffic via rules
- Works on ports, IP, protocol
- Applied to subnet/VM

One-liner:

NSG is a firewall that controls inbound and outbound traffic.

What is RBAC?

Answer :

- Role-based access control
- Assigns permissions
- Based on roles
- Enhances security

One-liner:

RBAC controls who can access Azure resources.

What is Load Balancer?

Answer :

- Distributes traffic
- Improves availability
- Supports high performance
- Works at Layer 4

One-liner:

Load Balancer distributes incoming traffic across resources.

What is VPN Gateway?

Answer :

- Connects on-prem to Azure
- Supports Site-to-Site
- Supports Point-to-Site
- Encrypted connection

One-liner:

VPN Gateway securely connects on-premises to Azure.

What is Availability Set?

Answer :

- Protects against failures
- Uses fault & update domains
- Improves uptime
- Used with VMs

One-liner:

Availability Set ensures VM uptime during failures and maintenance.

What is Azure Scale Set?

Answer :

- Manages group of identical VMs
- Supports auto-scaling
- Load balanced by default
- High availability

One-liner:

Scale Set automatically manages and scales multiple VMs.

What is Azure Application Gateway?

Answer :

- Layer 7 load balancer
- Supports URL-based routing
- Web Application Firewall (WAF)
- SSL termination

One-liner:

Application Gateway is a Layer 7 load balancer with advanced routing.

Difference between Load Balancer vs Application Gateway

Answer :

- Load Balancer → Layer 4 (TCP/UDP)
- App Gateway → Layer 7 (HTTP/HTTPS)
- LB → basic traffic distribution
- App Gateway → smart routing

One-liner:

Load Balancer works at L4, App Gateway works at L7 with intelligent routing.

What is Azure Functions?

Answer :

- Serverless compute

- Event-driven
- Pay per execution
- Supports multiple triggers

One-liner:

Azure Functions runs code without managing servers.

What is Azure Key Vault?

Answer :

- Stores secrets securely
- Manages keys & certificates
- Access controlled via RBAC
- Used in DevOps pipelines

One-liner:

Key Vault securely stores secrets, keys, and certificates.

What is Azure Site Recovery (ASR)?

Answer :

- Disaster recovery solution
- Replicates workloads
- Failover support
- Business continuity

One-liner:

ASR ensures disaster recovery by replicating workloads.

What is Azure Bastion?

Answer :

- Secure RDP/SSH access
- No public IP required
- Browser-based access
- Improves security

One-liner:

Bastion provides secure VM access without exposing public IP.

What is Azure Front Door?

Answer :

- Global load balancer
- Layer 7 routing
- Low latency routing
- CDN + WAF integration

One-liner:

Front Door routes global traffic with low latency and high availability.

What is Azure Traffic Manager?

Answer :

- DNS-based load balancing
- Routes based on priority/latency

- Supports multiple endpoints
- Global distribution

One-liner:

Traffic Manager distributes traffic using DNS routing.

What is Managed Identity?

Answer :

- Identity for Azure resources
- No credentials needed
- Used to access services securely
- Integrated with Azure AD

One-liner:

Managed Identity allows secure access without storing credentials.

What is Azure Policy?

Answer :

- Enforces governance rules
- Ensures compliance
- Can deny or audit resources
- Works at scale

One-liner:

Azure Policy enforces rules and compliance in resources.

What is Private Endpoint?

Answer :

- Private IP access to services
- Avoids public internet
- Enhances security
- Works with VNet

One-liner:

Private Endpoint connects services privately within a VNet.

What is Service Endpoint?

Answer :

- Extends VNet to Azure services
- Secure connection
- Still uses public endpoint
- Simpler than Private Endpoint

One-liner:

Service Endpoint secures Azure services via VNet integration.

Difference: Service Endpoint vs Private Endpoint

Answer :

- Service Endpoint → Public IP
- Private Endpoint → Private IP
- Private Endpoint more secure

- Service Endpoint easier setup

One-liner:

Private Endpoint is more secure than Service Endpoint.

. What is Azure Landing Zone?

Answer :

- Predefined cloud environment setup
- Follows best practices
- Includes networking, security, governance
- Built using ARM/Bicep/Terraform

One-liner:

Landing Zone is a secure, scalable foundation for Azure environments.

What is Zero Trust Security in Azure?

Answer :

- Never trust, always verify
- Identity-based access
- Least privilege principle
- Continuous monitoring

One-liner:

Zero Trust means verify everything before granting access.

Event Hub vs Service Bus

Answer :

- Event Hub → streaming data
- Service Bus → messaging system
- Event Hub → high throughput
- Service Bus → guaranteed delivery

One-liner:

Event Hub is for streaming, Service Bus is for messaging.

What is Azure Firewall?

Answer :

- Managed firewall service
- Filters traffic
- Supports rules & logging
- High availability

One-liner:

Azure Firewall protects resources by filtering network traffic.

What is Azure Disk Types?

Answer :

- Standard HDD → Low cost
- Standard SSD → Balanced
- Premium SSD → High performance
- Ultra Disk → Extreme performance

One-liner:

Azure offers multiple disk types based on performance and cost.

What is Azure Availability Zone vs Region Pair?

Answer :

- AZ → within same region
- Region Pair → different regions
- AZ → low latency
- Region Pair → disaster recovery

One-liner:

AZ is local redundancy, Region Pair is global disaster recovery.

What is Azure Tenant?

Answer :

- Instance of Azure AD
- Represents organization
- Contains users & apps
- Linked to subscriptions

One-liner:

Tenant is the identity boundary for an organization in Azure.

What is ExpressRoute?

Answer :

- Private connection to Azure
- Faster & secure than VPN
- No internet exposure
- Used by enterprises

One-liner:

ExpressRoute provides private, high-speed connection to Azure.

How do you design a highly available architecture in Azure?

Answer :

- Use **Availability Zones**
- Deploy multiple VMs / instances
- Add **Load Balancer / Application Gateway**
- Use **Auto Scaling**
- Implement backup & disaster recovery

One-liner:

High availability is achieved using redundancy, load balancing, and failover mechanisms.

How do you design a scalable architecture?

Answer :

- Use **VM Scale Sets / App Service scaling**
- Enable auto-scaling based on metrics
- Use stateless architecture
- Add caching (Redis)
- Use CDN

One-liner:

Scalability is achieved by auto-scaling and stateless architecture design.

How do you secure Azure resources?

Answer :

- Use **RBAC (least privilege)**
- Enable **NSG + Firewall**
- Use **Private Endpoints**
- Store secrets in **Key Vault**
- Enable monitoring & alerts

One-liner:

Security is ensured using RBAC, network isolation, and secret management.

How do you optimize cost in Azure?

Answer :

- Use **Reserved Instances**
- Enable auto-scaling
- Shut down unused VMs
- Use **Spot VMs**
- Monitor via cost management

One-liner:

Cost optimization is done using scaling, reservations, and monitoring usage.

How do you connect on-premises to Azure?

Answer :

- Use **VPN Gateway** (quick setup)
- Use **ExpressRoute** (private connection)
- Configure VNet integration
- Secure with NSG & firewall

One-liner:

Hybrid connectivity is achieved via VPN or ExpressRoute.

How do you handle disaster recovery?

Answer :

- Use **Azure Site Recovery**
- Replicate to another region
- Use backups
- Test failover regularly
- Define RTO & RPO

One-liner:

Disaster recovery ensures business continuity via replication and failover.

Difference: Azure VM vs App Service

Answer :

- VM → Full control
- App Service → Managed platform
- VM → OS level access
- App Service → No OS management

One-liner:

VM gives full control, App Service is fully managed

What is fault domain vs update domain?**Answer :**

- Fault domain → Hardware failure isolation
- Update domain → Maintenance isolation
- Used in Availability Set
- Prevents downtime

One-liner:

Fault domain = hardware isolation, update domain = maintenance isolation.

How do you reduce latency globally?**Answer :**

- Use **Azure Front Door**
- Use **CDN for static content**
- Deploy apps in multiple regions
- Use Traffic Manager

One-liner:

Global latency is reduced using CDN and global load balancing.

How do you implement Zero Downtime Deployment?**Answer :**

- Use **deployment slots (App Service)**
- Blue-Green deployment
- Rolling updates
- Health checks before swap

One-liner:

Zero downtime is achieved using blue-green or rolling deployments.

How do you enforce governance in Azure?**Answer :**

- Use **Azure Policy**
- Use **Blueprints**
- Tagging strategy
- Role-based access

One-liner:

Governance is enforced using policies, roles, and standards.

Hub-Spoke Architecture in Azure?**Answer :**

- Hub → central services (firewall, VPN)
- Spoke → application VNets
- Improves security
- Scalable design

One-liner:

Hub-Spoke centralizes networking and improves security.

What is Azure Data Factory?

Answer :

- ETL/ELT service
- Data integration
- Supports pipelines
- Connects multiple sources

One-liner:

Data Factory is used for data integration and transformation.

How do you choose between Azure services?

Answer :

- Based on workload
- Cost vs performance
- Scalability needs
- Management overhead
- Security requirements

One-liner:

Service selection depends on workload, cost, and scalability needs.

What is the biggest challenge in cloud?

Answer :

- Cost management
- Security
- Complexity
- Skill gap
- Governance

One-liner:

Managing cost, security, and complexity is the biggest challenge.

What is your approach to troubleshooting in Azure?

Answer :

- Check logs (Monitor)
- Validate configuration
- Check network rules
- Use diagnostics
- Reproduce issue

One-liner:

Troubleshooting involves logs, configuration checks, and diagnostics.

How do you handle production issues?

Answer :

- Identify root cause
- Check monitoring alerts
- Apply quick fix
- Communicate with team
- Prevent future issues

One-liner:

Handle production issues by fixing fast and preventing recurrence.

How do you design a secure network architecture in Azure?

Answer :

- Use **VNet with subnets (segmentation)**
- Apply **NSG rules**
- Use **Azure Firewall** for centralized control
- Implement **Private Endpoints**
- Enable DDoS protection

One-liner:

Secure architecture uses network isolation, firewalls, and private access.

. Difference: Azure Front Door vs Traffic Manager

Answer :

- Front Door → Layer 7 (HTTP/HTTPS)
- Traffic Manager → DNS-based
- Front Door → real-time routing
- Traffic Manager → DNS resolution-based

One-liner:

Front Door is L7 routing, Traffic Manager is DNS-based routing.

How do you handle secrets in CI/CD?

Answer:

- Use **Azure Key Vault**
- Integrate with pipeline
- Avoid hardcoding
- Use Managed Identity
- Rotate secrets

One-liner:

Secrets are managed securely using Key Vault and pipeline integration.

What is RTO & RPO?

Answer:

- RTO → Recovery Time Objective
- RPO → Recovery Point Objective
- Used in disaster recovery
- Defines downtime & data loss

One-liner:

RTO = recovery time, RPO = acceptable data loss.

What is Network Peering?

Answer :

- Connects VNets
- Private communication
- Low latency
- No internet involved

One-liner:

VNet peering connects networks privately.

What is Azure Availability Set vs Scale Set?

Answer :

- Availability Set → fault isolation
- Scale Set → scaling + management
- Scale Set supports auto scaling
- Availability Set does not

One-liner:

Availability Set ensures uptime, Scale Set ensures scaling.

What happens if NSG and Firewall conflict?

Answer :

- Both rules apply
- Most restrictive rule wins
- Traffic evaluated step by step
- Logging helps debugging

One-liner:

The most restrictive rule is applied in case of conflict

What is your deployment strategy preference?

Answer :

- Blue-Green → safer
- Rolling → gradual
- Canary → controlled release
- Depends on use case

One-liner:

Choose deployment strategy based on risk and application type.

How do you ensure compliance in Azure?

Answer :

- Use Azure Policy
- Audit logs
- Role-based access
- Security Center
- Regular reviews

One-liner:

Compliance is ensured using policies, auditing, and access control.